

Snort 2.9.20 Network Intrusion Detection Setup and Rule Configuration Report

Network IDS laboratory implementation with custom detection rules and testing evidence

1. Introduction

Snort is an open-source Network Intrusion Detection System (NIDS) used to inspect network traffic and generate alerts when traffic matches configured detection rules.

2. Objective

The objective is to monitor the protected network and detect ICMP activity, abnormal ICMP traffic, and TCP SYN-based reconnaissance.

3. Snort Version Verification

The first screenshot verifies Snort 2.9.20 GRE (Build 82) and its packet-capture and supporting libraries.

4. Promiscuous Mode

The second screenshot shows promiscuous mode being enabled on ens33 using `ip link set ens33 promisc on`.

5. Network Configuration

The protected network is configured with `HOME_NET 192.168.244.1/24`, `EXTERNAL_NET any`, and `DNS_SERVERS $HOME_NET`.

6. Local IDS Rules

Custom local rules are configured for ICMP Echo Requests, ICMP Echo Replies, large ICMP packets, high-rate ICMP traffic, and TCP SYN port scanning.

7. ICMP Detection

SID 1000001 detects ICMP Echo Requests. SID 1000002 detects Echo Replies. SID 1000003 detects ICMP payloads larger than 800 bytes. SID 1000004 detects repeated ICMP Echo Requests using a source-based threshold of 10 packets within 5 seconds.

8. TCP SYN Port-Scan Detection

SID 1000010 detects TCP SYN traffic using flags:S and a source-based threshold of 20 packets within 5 seconds.

9. Configuration Validation

The configuration can be tested with: `snort -T -c /etc/snort/snort.conf`

10. Running Snort

A typical command is: `snort -i ens33 -c /etc/snort/snort.conf`

11. IDS Testing and Alert Detection

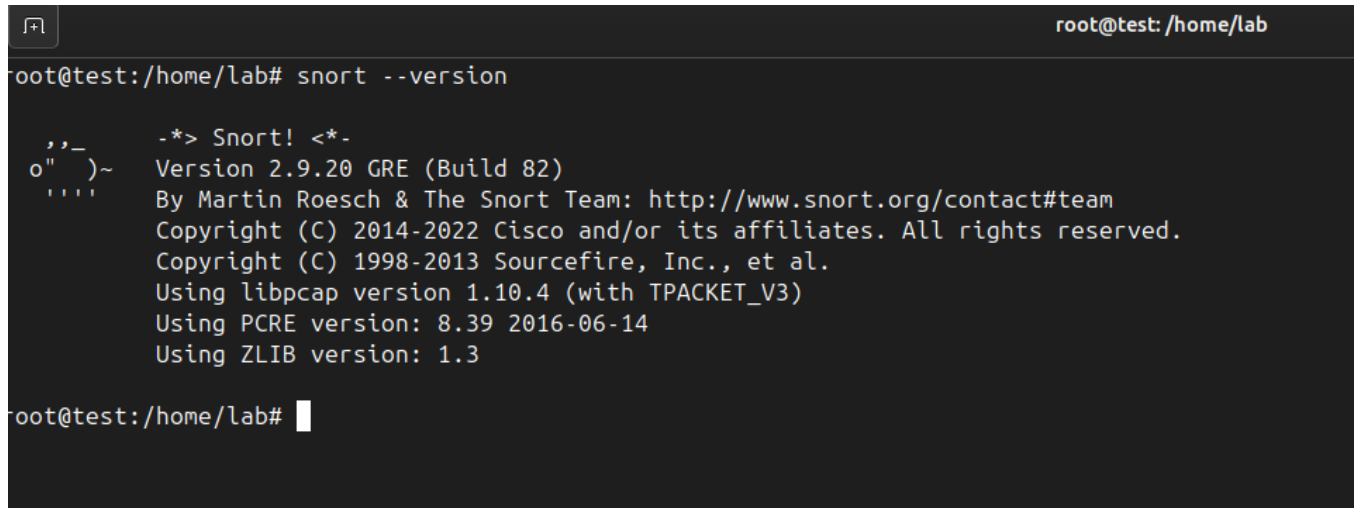
The final screenshot demonstrates a live test. Kali Linux runs `nmap -sS 192.168.244.1/24` while Ubuntu runs Snort. The Snort console shows repeated "Possible SYN port scan" alerts and inbound connection alerts, demonstrating that the IDS is detecting the SYN scan traffic.

12. Test Result

The test successfully demonstrates the IDS detection process: Nmap generates SYN traffic, Snort inspects the packets, the custom rules match the traffic, and security alerts are generated.

13. Conclusion

The project demonstrates a functional Snort-based Network IDS with custom detection rules and practical scan-detection evidence. It provides a foundation for further network security monitoring and SIEM integration.

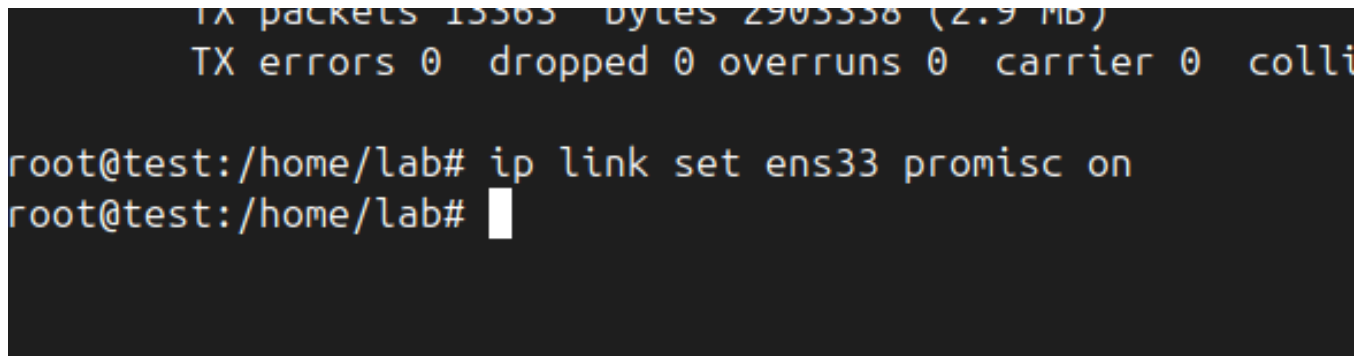
A terminal window with a dark background and light-colored text. The prompt is 'root@test:/home/lab#'. The command 'snort --version' has been executed, resulting in a multi-line output block. The output includes a decorative ASCII art header, the version number '2.9.20 GRE (Build 82)', the authors 'By Martin Roesch & The Snort Team', copyright information for Cisco and Sourcefire, and the versions of dependencies: libpcap 1.10.4, PCRE 8.39, and ZLIB 1.3.

```
root@test:/home/lab# snort --version

,,_      -*> Snort! <*-
o" )~    Version 2.9.20 GRE (Build 82)
' ' '    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
        Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
        Copyright (C) 1998-2013 Sourcefire, Inc., et al.
        Using libpcap version 1.10.4 (with TPACKET_V3)
        Using PCRE version: 8.39 2016-06-14
        Using ZLIB version: 1.3

root@test:/home/lab#
```

Figure 1: Snort 2.9.20 version verification

A terminal window showing network statistics and a command to enable promiscuous mode. The statistics line shows 'TX packets 13363' and 'bytes 2903338 (2.9 MB)'. Below it, another line shows 'TX errors 0', 'dropped 0', 'overruns 0', 'carrier 0', and 'colli'. The prompt is 'root@test:/home/lab#'. The command 'ip link set ens33 promisc on' has been entered, and the prompt is shown again below it.

```
TX packets 13363  bytes 2903338 (2.9 MB)
TX errors 0  dropped 0 overruns 0  carrier 0  colli

root@test:/home/lab# ip link set ens33 promisc on
root@test:/home/lab#
```

Figure 2: Enabling promiscuous mode on ens33

```

# Setup the network addresses you are protecting
#
# Note to Debian users: this value is overridden when starting
# up the Snort daemon through the init.d script by the
# value of DEBIAN_SNORT_HOME_NET s defined in the
# /etc/snort/snort.debian.conf configuration file
#
ipvar HOME_NET 192.168.244.1/24

# Set up the external network addresses. Leave as "any" in most
ipvar EXTERNAL_NET any
# If HOME_NET is defined as something other than "any", altern
# use this definition if you do not want to detect attacks from
# IP addresses:
#ipvar EXTERNAL_NET !$HOME_NET

# List of DNS servers on your network
ipvar DNS_SERVERS $HOME_NET

```

Figure 3: HOME_NET, EXTERNAL_NET, and DNS_SERVERS configuration

```

# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures. Put your local
# additions here.
# -----

# =====
# ICMP / Ping detection
# =====

# Any inbound ping
alert icmp any any -> $HOME_NET any (msg:"ICMP ping detected"; itype:8; sid:1000001; rev:1;)

# Ping reply leaving your network (confirms a host responded)
alert icmp $HOME_NET any -> any any (msg:"ICMP ping reply sent"; itype:0; sid:1000002; rev:1;)

# Large/oversized ping packet - classic ping-flood or ping-of-death style probe
alert icmp any any -> $HOME_NET any (msg:"Large ICMP packet - possible ping flood"; dsize:>800; sid:1000003; rev:1;)

# Rapid repeated pings from the same source (basic ICMP flood / sweep detection)
alert icmp any any -> $HOME_NET any (msg:"Possible ICMP flood - high ping rate"; itype:8; detection_filter:track by_src, count 10, seconds 5; sid:1000004; rev:1;)

# =====
# TCP / Port scan detection
# =====

# SYN scan / stealth scan attempt (SYN flag only, no ACK - typical of nmap -sS)
alert tcp any any -> $HOME_NET any (msg:"Possible SYN port scan"; flags:S; detection_filter:track by_src, count 20, seconds 5; sid:1000010; rev:1;)

# NULL scan (no flags set - used to evade basic detection)

```

Figure 4: Custom ICMP and TCP detection rules

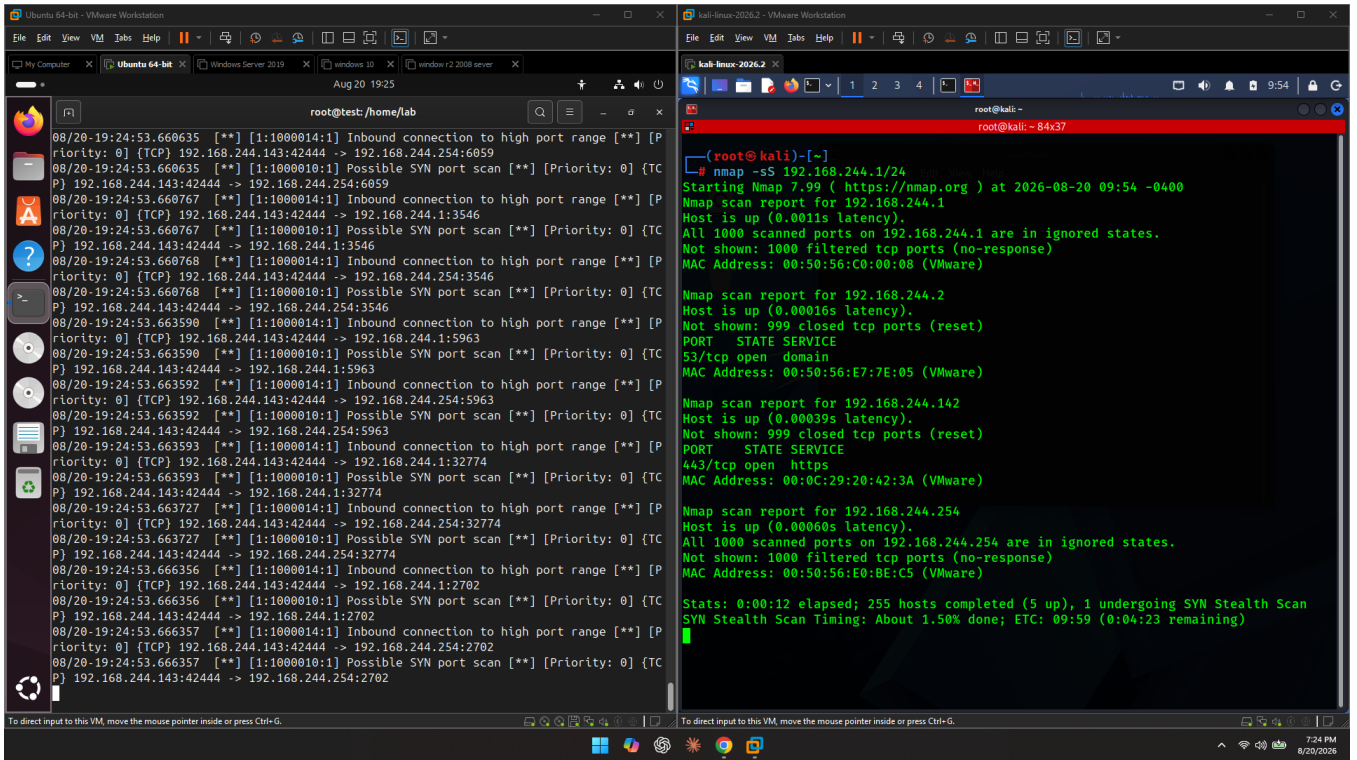


Figure 5: Nmap SYN scan from Kali Linux and Snort alerts on Ubuntu

Detection Rule Summary

SID	Protocol	Detection	Threshold
1000001	ICMP	Echo Request	Type 8
1000002	ICMP	Echo Reply	Type 0
1000003	ICMP	Large ICMP	Payload > 800 bytes
1000004	ICMP	High-rate ICMP	10 / 5 sec / source
1000010	TCP	Possible SYN scan	20 / 5 sec / source